MINI-PILOT PROTOCOL

AI Data Centre Facilities Assurance Framework: pilot pack

Version 1.0

A protocol for running a small, authorised AIDCF research assessment of one bounded AI Capability Instance at Assurance Level AL3

Artifact ID: AIDCF-PIL-001 Short code: PIL Artifact type: Pilot protocol (informative; applies AIDCF-009, AIDCF-010, AIDCF-011 and AIDCF-012 without amending them) Status: Draft for controlled review Author: Siva Sethumadhavan

> **Pilot status: pack ready; no authorised pilot has taken place.** This protocol is a plan. It records no pilot, host, sponsor, approval or result. Whitepaper §29 status PILOT-001: NOT AUTHORISED / NOT EXECUTED.

Document control

Field	Value
Document ID	AIDCF-PIL-001
Title	AIDCF Mini-Pilot Protocol
Version	1.0
Date	2026-10-11
Status	Draft for controlled review
Author	Siva Sethumadhavan
Applies	AIDCF-009 Part A v1.1 (lifecycle, stage gates SG0–SG7, rules of engagement B14, workpapers WP-01 to WP-20); AIDCF-010 v1.1 (mandatory claim types, gate roll-up, decision derivation); AIDCF-011 v1.1 (Assurance Levels, independence, templates); AIDCF-012 v1.1 (report classes, controlled wording, classification); AIDCF-008 v1.1 (evidence floors); AIDCF-007A registry v1.1
Companion records	AIDCF-PIL-002 Pilot Authorisation Template; AIDCF-PIL-003 Evidence Request List Template; AIDCF-PIL-004 Pilot Result Record Template; AIDCF-PIL-005 Lessons Learned Template
Machine-readable files	`pilots/pilot-framework/starter-profile.csv`; `pilots/pilot-authorisation/evidence-request-list.csv`; `pilots/pilot-results/control-claim-log-template.csv` (all generated by `pilots/pilot-framework/tools/build_starter_profile.py`); `pilots/pilot-results/pilot-metrics-template.csv`
Requirement IDs	PIL-REQ-001 to PIL-REQ-016

Revision history

Version	Date	Summary
1.0	2026-10-11	First issue.

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1. Purpose and status

AIDCF has been exercised only on synthetic material (AIDCF-015) and self-validation. Whitepaper §30 lists "Real capability pilot" and "Inter-rater reproducibility" as not completed. This protocol defines the smallest assessment that is still a genuine AIDCF assessment, so that a host organisation can authorise it, a competent team can run it, and the results can be fed back into the framework.

The protocol is deliberately conservative:

- one bounded Capability Instance;
- target Assurance Level AL3 (DESIGN and IMPLEMENTATION claims), with AL4 as an option where operating records exist;
- desk review, walkthrough and configuration inspection only; no test, switching, isolation or other intervention by the pilot team;
- a fixed Starter Profile of 58 controls generated from the AIDCF-007A registry;
- seven mapped critical gates plus the two meta-gates in scope; the remaining six gates reported NOT_ASSESSED with their decision effect.

The pilot is a real assessment with a real, bounded decision, and it is also a research instrument. The two purposes are kept separate: the Assurance Decision follows AIDCF-010 and AIDCF-012 without modification, and research measurements (§12) never change a result.

The protocol does not amend any AIDCF standard. Where applying a standard needed an interpretation, the interpretation is stated (§6.5, §6.6, §7.2) and is itself a research item for the framework owner.

PIL-REQ-001 A pilot shall not start fieldwork until AIDCF-PIL-002 has been signed by every party it names and SG2 has been recorded as PROCEED (AIDCF-009 A6).

PIL-REQ-002 No statement that a pilot has taken place, and no pilot result, shall be published in this repository unless the signed pilot records exist and the host has approved publication under AIDCF-PIL-002 §12 (README "Pilot programme"; GOVERNANCE.md §15).

2. Objectives and research questions

2.1 Objectives

ID	Objective
O1	Demonstrate that a bounded AIDCF assessment can be authorised, planned, executed, reviewed and decided at a real host using AIDCF-009 Part A as written.
O2	Measure the practical cost of AIDCF at AL3: assessor and host effort per control-claim, per evidence request and per gate.
O3	Measure evidence availability against the registry evidence profiles, including availability of evidence that would be needed at AL4 and AL5.

ID	Objective
O4	Measure how often results are UNKNOWN, why (U1–U10) and with what severity (US1–US5).
O5	Measure inter-rater agreement on a defined sample of control results and on all in-scope Gate Results.
O6	Find defects in controls, assessment criteria, gate criteria, evidence profiles, vocabularies and templates, and usability problems in the method.
O7	Obtain the named decision user's view of whether the bounded output was useful.

2.2 Research questions informed

The pilot informs, and cannot by itself answer, the following README research questions. One pilot is a single case; it produces observations and candidate defects, not generalisable findings.

RQ	What this pilot contributes	Measures (§12)
RQ1 Subject definition	Independent first drafts of the Capability Instance, Service Outcomes and boundary by two assessors before reconciliation.	PM-11
RQ2 Cross-domain completeness	Findings that cannot be mapped to any control (orphan findings).	PM-14, PM-15
RQ3 Control-universe validity	Defects in Starter Profile control requirements and criteria.	PM-14
RQ4 Dependency discovery	Dependencies found under CG-03 that the host's own records did not show.	PM-16
RQ6 Evidence quality	Achieved sufficiency against profile floors; feasibility of mandatory items.	PM-03, PM-04, PM-07
RQ7 UNKNOWN discipline	UNKNOWN rate, types, severities and their decision effect.	PM-05, PM-06
RQ8 Critical-gate effectiveness	Cases where gate results and scores point in different directions.	PM-17
RQ10 Inter-rater reliability	Control-result and Gate Result agreement and disagreement causes.	PM-08, PM-09, PM-10
RQ11 Machine traceability	Completeness of the AMH-REQ-017 trace for the Decision Recommendation.	PM-12
RQ13 Decision usefulness	Structured rating by the named decision user.	PM-18
RQ15 Scalability	Effort and elapsed time against the planning assumptions in §9.	PM-01, PM-02, PM-19

RQ5, RQ9, RQ12, RQ14 and RQ16 are out of scope: they need EFFECTIVENESS/RESILIENCE claims, scored populations across sites, change events over time, authorised economic data or AI Trust Graph federation, none of which an AL3 mini-pilot provides.

3. Eligible hosts and minimum conditions

3.1 Eligible hosts

Suitable hosts operate one AI or HPC capability whose supporting infrastructure they can describe and give evidence for, for example:

- a university or research-institute HPC centre operating a GPU partition;
- a laboratory or departmental AI training/inference cluster in an owned or shared machine room;
- a research-computing service operated by a public body or consortium;
- a commercial operator willing to run a research assessment of one hall or pod under the same terms.

Two boundary variants are common and both are acceptable:

Variant	Description	Consequence
A. Integrated	The host (possibly together with its estates or facilities function) owns and operates the IT capability and the power, cooling and control systems serving it.	All seven mapped gates are in scope as set out in §8.
B. Tenant in a facility	Power and/or cooling are provided by a separate party (campus estates, colocation provider) under an agreement.	CG-05 or CG-06 may be APPROVED_NA only if the registry APPROVED_NA conditions are met in full: the entire electrical path to the rack interface (CG-05) or the entire heat-removal path including in-rack and CDU elements (CG-06) is outside the boundary and evaluated as an external dependency under CG-03, with evidence rights; for CG-05, CG-03 must then PASS for that dependency. Otherwise the provider must be party to the authorisation and the gate stays in scope.

3.2 Minimum conditions

A host is eligible only if every condition below is met and evidenced in AIDCF-PIL-002. These restate the README "Pilot programme" list and GOVERNANCE.md §15 in operational form.

#	Condition	Evidence in AIDCF-PIL-002
C1	**Named sponsor** with authority over the decision question, the scope and the resources, able to commit host staff time.	§2 sponsor authorisation
C2	**Capability owner** accountable for the Capability Instance and its Service Outcomes (AIDCF-009 A4), and, in Variant B, the owner of each provided service.	§2, §3
C3	**Decision question** that a named decision user actually needs answered, stated in the form required by AIDCF-009 §0.5, with a prohibited interpretation.	§4
C4	**Decision authority** independent of the assessment team, with authority for the decision question (GC-13-6).	§2, §14
C5	**Safety authority** for the site who can approve walkthroughs, inspections and configuration extraction and can stop any pilot activity (AMH-REQ-031).	§8
C6	**Evidence access** to the records listed in AIDCF-PIL-003, including configuration extraction executed or witnessed by the assessor, under agreed classification and custody.	§6, §9, §10
C7	**Independent reviewer** with no preparation role and no self-review threat (AMH-REQ-011, AMH-REQ-012; Part B AC-5).	§14

#	Condition	Evidence in AIDCF-PIL-002
C8	**Competent assessors** covering the disciplines in §10, with conflicts declared.	§13
C9	**Classification, retention and publication terms** agreed, including anonymisation rules for research outputs.	§9, §11, §12
C10	**Legal review** of the authorisation by the host's legal adviser before signature.	Front matter

Where any condition cannot be met, the engagement is declined at DT-01 (AIDCF-009 Part B) or returned to the sponsor before SG0. An eligible but under-resourced host can instead run an AL1 readiness review (AIDCF-011 Part IV), which issues no Assurance Decision and is not a pilot under this protocol.

PIL-REQ-003 The sponsor, capability owner, decision authority, safety authority and independent reviewer shall be five named roles; the decision authority and the independent reviewer shall be different persons from each other and from every preparer (AIDCF-009 A4). The sponsor and capability owner may be the same person only if that is recorded with the reason.

4. Recommended scope

4.1 Capability Instance

Scope one bounded Capability Instance: one AI training or inference cluster (or one partition of a larger HPC system with its own scheduler boundary) together with the supporting power, cooling, network and control systems on which its Service Outcomes depend. A typical boundary is:

Layer	Typical content inside the boundary
Capability	The cluster or partition, its scheduler, its Service Outcomes (availability, performance, integrity, recovery) and the decision question.
Compute, storage, network	Accelerator nodes, the storage tiers holding training data, checkpoints and models for the cluster, the compute and storage fabrics, management and out-of-band networks.
Power	From the point of common coupling or campus supply point to the rack power interface: transformers, switchgear, UPS/BESS, generation and fuel where present, distribution, EPMS.
Cooling	Heat-removal path from the accelerators to heat rejection: in-rack and CDU elements, secondary and facility water loops, chillers, dry coolers or towers, air systems, water supply.
Control systems	BMS, EPMS, CDU controllers, PLC/SCADA and their networks where they can change the state of in-scope power or cooling.
Dependencies	Utility, telecom, cloud and service providers, OEM support and sub-tier suppliers on which the above depend (recorded under CG-03).

Shared campus plant that also serves other loads stays inside the boundary where it is a Required Function of the Service Outcomes; the other loads are recorded as concentration and common-mode context under CG-03, not assessed.

4.2 Operating states and period

The DESIGN and IMPLEMENTATION claims are made for the operating states named in each control's evidence profile (`stateCoverage`), typically NORMAL, MAINTENANCE, DEGRADED, EMERGENCY and RECOVERY. At AL3 these states are assessed through design intent and as-built configuration; behaviour in those states over

time is an OPERATION claim and is outside AL3. The temporal boundary is the assessment date, with evidence currency per profile `periodCoverage`.

4.3 Exclusions

Every exclusion is recorded in WP-03 with its rationale, affected Service Outcomes and decision effect (AMH-REQ-020). Default exclusions:

- other clusters, tenants and workloads sharing the facility (context only);
- application, model and data-governance assurance (outside AIDCF scope; AI Trust Graph domain);
- the 302 registry controls outside the Starter Profile (§6.6);
- all EFFECTIVENESS and RESILIENCE claims, and OPERATION claims unless AL4 is chosen (§5).

5. Target Assurance Level

5.1 Recommendation: AL3

The recommended target is **AL3 Implementation** (AIDCF-011 Part IV): DESIGN and IMPLEMENTATION claim types, CP-2 evidence typical, decision states applying to design and implementation claims only.

Aspect	AL3 mini-pilot setting
Claim types concluded	DESIGN; IMPLEMENTATION (AIDCF-010 Part IV with T = AL3).
Methods	DOCUMENT_REVIEW, INQUIRY (corroborating only, AMH-REQ-023), INSPECTION (non-intrusive), OBSERVATION (walkthroughs), RECONCILIATION, SAMPLING, ANALYSIS.
Methods excluded	TEST, REPERFORMANCE of operational procedures and SCENARIO_EXERCISE by the pilot team. Existing test, commissioning and exercise records are reviewed as documents. Observing a host-run tabletop walkthrough is OBSERVATION and is permitted where the safety authority agrees.
Configuration inspection	Read-only extraction executed by the host's authorised operator and witnessed by the assessor, or executed by the assessor with read-only credentials issued for the pilot; extraction context recorded (AIDCF-009 B8 limit).
Disruptive activity	None. AMH-REQ-026 and AMH-REQ-029 apply in full.
Claim depth stated in results	Every Gate Result states "at IMPLEMENTATION" or "at DESIGN" (SDF-REQ-010); the decision asserts nothing deeper.

AL3 is recommended because it is the deepest level that a research team can reach without operating records over a period, without any test authority, and with a fieldwork footprint (desk work plus a few days of walkthroughs) that a host can accept.

5.2 Option: AL4

Where the host holds at least twelve months of operating records for the in-scope systems (EPMS and BMS trends, alarm and event logs, maintenance records, incident tickets, scheduler availability data) and agrees to share them, the sponsor may set T = AL4 at SG1. Then, by L = min(T, r), every Starter Profile control with r = AL4 or AL5 (31 of the 58) also carries OPERATION as a mandatory claim type, adding 31 control-claim assessments (138 in total), and the effective gate evidence floor rises to the gate floor for gates whose CRITICAL controls are evaluated at OPERATION depth (AIDCF-010 Part VIII). The choice shall be frozen in the plan before results are

recorded (SDF-REQ-008). The generated `mandatoryClaimTypesAtAL3` column then no longer applies; the plan records the AL4 set.

AL5 is not offered: EFFECTIVENESS and RESILIENCE claims need tests or exercises under AMH-REQ-029, which are out of keeping with a research pilot.

PIL-REQ-004 The target Assurance Level shall be AL3 or AL4, recorded in WP-01 and frozen at SG1; it shall not be changed after the first control result is recorded.

6. Starter Profile

6.1 Content

The Starter Profile is `pilots/pilot-framework/starter-profile.csv`, generated by `pilots/pilot-framework/tools/build_starter_profile.py` from the AIDCF-007A registry v1.1. It contains 58 controls:

Domain	Controls	Domain	Controls
CAP	7	STO	5
DEP	8	NET	3
POW	10	OPS	1
THM	9	RES	4
CMP	5	SUP	1
SEC	3	EVA	2

At AL3, 9 controls (registry r = AL2) carry DESIGN only and 49 carry DESIGN and IMPLEMENTATION: **107 mandatory control-claim assessments**.

Columns: `controlId`, `domain`, `title`, `claimType` (registry profile claim type p), `requiredAssuranceLevel` (registry r), `mandatoryClaimTypesAtAL3` (computed), `criticalGates` (registry value, all gates), `selectionReason` (rule and in-scope gate roles).

6.2 Selection rule

In-scope mapped gates G = CG-01, CG-02, CG-03, CG-05, CG-06, CG-07, CG-10.

Rule	Selects	Count	Reason
R1	Every control holding a CRITICAL mapping to any gate in G.	50	Under the AIDCF-010 roll-up rule a gate cannot PASS while any CRITICAL control is NOT_ASSESSED at a mandatory claim type. Selecting only part of a gate's CRITICAL set would make that gate UNKNOWN by construction.
R2	Every AI-layer control (CMP, STO, NET) not selected by R1 that holds a SUPPORTING mapping to CG-01, or SUPPORTING mappings to two or more gates in G.	8	CG-01 SUPPORTING AI controls (accelerator architecture, usable accelerator capacity, training-data tier) define what the AI capability is. Multi-gate AI controls (rack-scale composition, spares and RMA, path documentation, time and core services, replication) sit where the AI layer meets power, cooling, dependency and recovery.

R1 selects all CRITICAL controls of the gates named in the pilot brief (CG-01, CG-05, CG-06, CG-07, CG-10: 39 controls), all nine CRITICAL controls of CG-03 (dependency and common-mode visibility: asset-to-function,

upstream, shared source, shared route, utility, telecom, cloud and service, boundary, sub-tier), and the two further CRITICAL controls of CG-02 (MCF-CAP-004 Architecture Linkage, MCF-EVA-010 Scope and Criteria; MCF-DEP-017 is already selected through CG-03).

CG-02 was added to the gates in the pilot brief because its UNKNOWN ceiling restricts "claims about systems whose reconciliation is incomplete" to the DESIGN claim type (registries/gates/critical-gates.csv). An AL3 pilot whose purpose is as-built reconciliation would otherwise be unable to state IMPLEMENTATION claims cleanly. The cost is two controls.

The brief asked for about 40 controls. The CRITICAL sets of the in-scope gates alone total 50, and any smaller set leaves at least one in-scope gate unable to PASS by construction. The profile is therefore 58 controls; a host that cannot support that size should remove a whole gate (and record it NOT_ASSESSED with its effect, §7.3) rather than part of one.

The generator enforces a post-condition that every CRITICAL control of every gate in G is selected, and its `--check` mode fails if the CSV files differ from a fresh build.

6.3 SUPPORTING controls outside the profile

Under the roll-up rule a SUPPORTING control that is PARTIAL, UNKNOWN or not assessed does not by itself change a Gate Result; only a NONCONFORMING SUPPORTING control needs a gate-level evaluation. SUPPORTING controls outside the profile are therefore recorded as out of pilot scope, listed among each gate's context in WP-14, and do not block PASS. Where fieldwork produces evidence that such a control is NONCONFORMING, the lead assessor adds it to scope through a recorded change at the current stage gate (AMH-REQ-005) and performs the gate-level evaluation.

6.4 Mandatory claim types

`mandatoryClaimTypesAtAL3` is computed with the AIDCF-010 Part IV rule L = min(T, r), T = AL3: r = AL2 gives DESIGN; r = AL3, AL4 or AL5 gives DESIGN;IMPLEMENTATION. Claim types above L (the profile claim type p where p is OPERATION, EFFECTIVENESS or RESILIENCE) are optional and are not assessed at AL3; they are neither results nor coverage gaps.

6.5 Interpretation used for evidence floors of shallower claim types

For a control whose profile claim type p is deeper than the claim type being assessed (for example the DESIGN claim of MCF-POW-011, whose profile is EFFECTIVENESS at S5), AIDCF-008 §31 states that "lower claim types are assessed with the patterns in §29". The pilot therefore applies the AIDCF-008 §29 minimum pattern for the shallower claim type, keeps the profile quality floor, and sets the sufficiency target at S3, the level AIDCF-008 §25 defines as supporting DESIGN or IMPLEMENTATION and the effective gate floor AIDCF-010 Part VIII sets at that depth. Where the profile claim type is itself DESIGN or IMPLEMENTATION, the profile floor applies unchanged even where it is above S3 (for example MCF-SEC-009 at S4); no floor is weakened (AMH-REQ-022, EVM-REQ-023).

The registry does not state this per claim type. The pilot records every case where the interpretation mattered to a result (PM-20) as evidence for the framework owner.

6.6 Coverage denominator

AIDCF-010 computes coverage over in-scope controls, and whitepaper §29.2 requires coverage to be "calculated against the approved applicable profile". The pilot computes decision coverage against the 107 mandatory control-claims of the Starter Profile. AIDCF-009 B4 also requires an applicability decision for all 360 controls. The 302 controls outside the profile are recorded in WP-04 as excluded from pilot scope by the approved profile (not NOT_APPLICABLE, which needs evidence that no Required Function depends on the control, AMH-REQ-021). They are disclosed as a scope exclusion in the decision and every report (AMH-REQ-020). For information only, the result record also shows coverage against all applicable controls, which will be below 50% by design; whether such an exclusion is a legitimate tailoring is a question for the framework owner (PM-20).

PIL-REQ-005 The Starter Profile used shall be the generated file at the commit recorded in WP-01, verified with `--check`; any change to it after SG1 shall be a recorded change approved at the current stage gate.

7. Gates in scope and the reachable decision

7.1 Gate scope

Gate	Name	Pilot scope	CRITICAL controls in profile	Effective evidence floor at AL3
CG-01	Capability and outcome definition	In scope	7 of 7	S3
CG-02	Boundary and architecture integrity	In scope	3 of 3	S3
CG-03	Critical dependency visibility	In scope	9 of 9	S3 (gate floor S4)
CG-04	Life safety and hazardous energy	NOT_ASSESSED	0 of 13	—
CG-05	Resilient electrical supply	In scope	8 of 8	S3 (gate floor S5)
CG-06	Thermal continuity	In scope	7 of 7	S3 (gate floor S5)
CG-07	Critical control-system integrity	In scope	6 of 6	S3 (gate floor S4)
CG-08	Compute, storage and network functional integrity	NOT_ASSESSED	1 of 12 (MCF-CMP-009, selected through R2; the gate is not evaluated)	—
CG-09	Failure containment	NOT_ASSESSED	0 of 9	—
CG-10	Recovery readiness	In scope	11 of 11	S3 (gate floor S5)
CG-11	Evidence sufficiency	In scope (meta-gate over the profile)	not mapped per control	S3 for DESIGN/IMPLEMENTATION claims
CG-12	Critical UNKNOWN	In scope (meta-gate over the profile)	not mapped per control	S3
CG-13	Responsibility and authority	NOT_ASSESSED	0 of 7	—

Gate	Name	Pilot scope	CRITICAL controls in profile	Effective evidence floor at AL3
CG-14	Change control	NOT_ASSESSED	0 of 7	—
CG-15	Integrated digital authority over infrastructure	NOT_ASSESSED	0 of 7	—

The effective floor follows AIDCF-010 Part VIII: where the deepest mandatory claim type of a gate's CRITICAL controls is DESIGN or IMPLEMENTATION, the effective floor is the lower of the gate floor and S3.

CG-04 is out of scope because life-safety conclusions need authorised engineering and safety records and often licensed judgement (AMH-REQ-015), and its CRITICAL controls are mostly OPERATION and EFFECTIVENESS controls whose AL3 claims add little. CG-04 cannot be APPROVED_NA where physical infrastructure lies inside the boundary. Being out of scope for assessment does not reduce the safety controls on the pilot's own activities (§9).

7.2 Gate criteria that need existing test records

Gate PASS needs every gate criterion met with evidence at or above the effective floor, as well as CONFORMING CRITICAL controls. Ten criteria of in-scope gates name TEST, SCENARIO_EXERCISE or REPERFORMANCE as the method: GC-05-4, GC-05-6, GC-05-7, GC-06-4, GC-06-6, GC-07-3, GC-07-7, GC-10-3, GC-10-4 and GC-10-5. The pilot performs no tests, so these criteria can be met only from the host's existing test, commissioning, acceptance or exercise records, reviewed as documents. Where such records do not exist or are out of currency, the criterion is unmet for lack of evidence and the gate is UNKNOWN (US4), not FAIL. AIDCF-PIL-003 marks these rows `MANDATORY_EXISTING_RECORDS_ONLY`.

The realistic expectation is therefore:

Gate	Realistic AL3 outcome
CG-01, CG-02, CG-03	PASS, FAIL or UNKNOWN at IMPLEMENTATION, on desk and reconciliation evidence alone.
CG-05, CG-06, CG-07, CG-10	PASS at IMPLEMENTATION only where current test and commissioning records exist for the criteria above; otherwise UNKNOWN. FAIL where evidence shows a criterion is not met.
CG-11, CG-12	Determined by the assessment team's own records; FAIL is possible and has whole-decision effect.

Whether gate criteria should be graded by claim depth, as gate floors are, is a research question the pilot records (PM-20).

7.3 Effect of NOT_ASSESSED gates

AIDCF-010 Part XII step 0 fixes the gate set as all fifteen unless APPROVED_NA. A gate that is applicable but not evaluated is NOT_ASSESSED and is treated as UNKNOWN in the derivation: its UNKNOWN ceiling applies (registries/gates/critical-gates.csv).

Gate	UNKNOWN ceiling applied when NOT_ASSESSED
CG-04	LIMITED for outcomes depending on the affected area; with the whole facility unassessed this covers every Service Outcome.
CG-08	LIMITED for the Service Outcomes delivered by the cluster, fabric or storage tier.

Gate	UNKNOWN ceiling applied when NOT_ASSESSED
CG-09	LIMITED for the decision as a whole; INCONCLUSIVE for RESILIENCE claims, which AL3 does not make.
CG-13	LIMITED.
CG-14	LIMITED.
CG-15	LIMITED; INCONCLUSIVE where the unverified authority reaches the scope of two or more other gates (US5).

For CG-15, the lead assessor shall apply the AIDCF-010 Part IX severity test whenever CG-07 fieldwork (for example MCF-SEC-009 zoning or MCF-SEC-005 BMS/EPMS security) shows an identity or remote-access path into control functions of two or more in-scope gates whose bounds have not been verified. Such a path is a US5 UNKNOWN and caps the decision at INCONCLUSIVE.

7.4 Reachable decision states

Because six applicable gates are NOT_ASSESSED, **the maximum permissible decision for any Service Outcome is LIMITED**, whatever the in-scope results are. This is by design: the pilot can say what is and is not supported within its boundary, at DESIGN and IMPLEMENTATION depth, and nothing more.

Decision state	Reachable?	How
REASONABLY_ASSURED	No	Needs all in-scope gates PASS or APPROVED_NA; six are NOT_ASSESSED.
CONDITIONALLY_ASSURED	No	Needs no gate FAIL or UNKNOWN affecting the outcomes; NOT_ASSESSED counts as UNKNOWN.
LIMITED	Yes, best case	Ceiling from the NOT_ASSESSED gates. The LIMITED wording (AIDCF-012 V.1) states the supported claims, for example "CG-01, CG-02 and CG-03 PASS at IMPLEMENTATION; CG-05 PASS at IMPLEMENTATION for power path A", and names the controlling Gate Results.
NOT_ASSURED	Yes	An in-scope gate FAILs with a NOT_ASSURED ceiling for the affected outcomes (for example CG-05 FAIL for outcomes on the affected electrical path).
INCONCLUSIVE	Yes	CG-01 FAIL (top claim cannot be framed); CG-11 or CG-12 FAIL; a US5 UNKNOWN (for example an undefined Service Outcome, or §7.3 CG-15 case); or profile coverage below 50%.

The useful output of the pilot is therefore not the decision state, which is predictable, but the Gate Results with their claim depth and controlling items, the UNKNOWN register, the reasons in AIDCF-010 step order, and the research measures. The sponsor and decision user shall be told this before SG0 (AIDCF-009 B1 escalation: a requested conclusion that exceeds what is achievable is returned to the sponsor).

PIL-REQ-006 The charter (WP-01) and AIDCF-PIL-002 §4 shall state that the maximum permissible decision is LIMITED under the scope in this section, and shall list the NOT_ASSESSED gates and their ceilings.

PIL-REQ-007 No pilot output shall describe a gate as passed, partially passed or conditionally passed unless its Gate Result is PASS, and every NOT_ASSESSED gate shall appear in the gate summary of every report with its ceiling (RAS-REQ-007, RAS-REQ-008).

8. Phases, stage gates and effort

8.1 Phase plan

Phases follow the AIDCF-009 §0.6 lifecycle and A6 stage gates; whitepaper §29.3 phases are shown for traceability.

Phase (§29.3)	AIDCF-009 lifecycle and procedures	Stage gate and approver	Pilot outputs
Authorise	Initiate: A1–A5, B1	SG0: sponsor	Signed AIDCF-PIL-002; WP-01 charter including PIL-REQ-006 statement; artifact-version manifest (AMH-REQ-005) including the Starter Profile commit.
Profile	Profile, Scope, Tailor: B2, B3, B4	SG1: sponsor and lead assessor	WP-02, WP-03 (Variant A or B), WP-04 applicability (58 in profile, 302 excluded by profile, §6.6); independent first drafts by two assessors for PM-11; target AL frozen.
Plan	Plan: B5, B14	SG2: lead assessor and safety authority	WP-05 plan with method map per control-claim; rules of engagement (AIDCF-PIL-002 §7); WP-06 evidence requests issued from AIDCF-PIL-003; WP-09 sampling plans; double-assessment sample drawn (§12.3).
Collect	Collect, Validate: B6–B12	SG3: lead assessor	Evidence manifest; WP-07 interviews; WP-08 walkthrough and inspection records; configuration extracts with context; UNKNOWN Records for missing, stale, refused or substituted items.
Assess; Gate and score	Assess, Analyse, Gate, Find: C1–C5	SG4: technical reviewers	WP-12 control results (107); WP-14 Gate Results (CG-01, CG-02, CG-03, CG-05, CG-06, CG-07, CG-10, CG-11, CG-12 evaluated; six NOT_ASSESSED); WP-15, WP-16, WP-17; Score Snapshot with coverage.
Decide (review)	Review: C6	SG5: independent reviewer	WP-18 covering every FAIL or UNKNOWN gate, every US4/US5 UNKNOWN and the wording (AMH-REQ-041); AMH-REQ-017 trace.
Decide; Report	Decide, Report: C7, C8	SG6: decision authority	WP-19 decision (or recorded decision not to issue one); R2 Technical Assessment Report; R5 Gate and Decision Package; R4 Evidence Package; R1 if the sponsor wants it.
Learn	Archive (SG7) and lessons	SG7: lead assessor	Immutable package; AIDCF-PIL-004 result record; AIDCF-PIL-005 lessons; metrics; anonymised summary if approved.

A stage gate has three outcomes: PROCEED, RETURN or STOP (AIDCF-009 A6). There is no conditional proceed; open items are carried as UNKNOWN Records or NOT_ASSESSED results.

8.2 Effort and duration: planning assumptions

The figures below are **planning assumptions for budgeting and authorisation, not facts or benchmarks**. No AIDCF assessment has been timed. Measuring them is one of the pilot's objectives (PM-01, PM-02, PM-19); the result record compares actuals with these assumptions.

Basis: 107 control-claims (58 DESIGN, 49 IMPLEMENTATION), 9 gate evaluations, 189 host-facing mandatory evidence request rows and 74 availability-only rows (AIDCF-PIL-003), a double-assessment sample of about 20% of control-claims, assumed per-item effort of 1–3 hours per DESIGN claim and 2–5 hours per IMPLEMENTATION claim including evidence review and workpaper.

Phase	Elapsed (weeks)	Pilot team (person-days)	Host (person-days)	Main driver of the range
Authorise (SG0)	2–6	3–6	2–5	Host legal and information-governance review.
Profile (SG1)	1–2	5–10	2–4	Maturity of the host's capability and boundary records; two independent drafts.
Plan (SG2)	1–2	3–6	1–3	Safety authority availability; extraction arrangements.
Collect (SG3)	3–6	8–15	10–25	Evidence location and retrieval; 2–4 days of walkthroughs on site.
Assess, gate (SG4)	3–5	30–80	2–5	107 control-claims at 1–5 h; 9 gates at 0.5–1.5 days each; double assessment; technical review.
Independent review (SG5)	1–2	3–8 (reviewer)	0–1	Number of FAIL/UNKNOWN gates and US4/US5 UNKNOWNs.
Decide, report (SG6)	1–2	4–8	1–2 (decision authority, sponsor)	Number of report classes.
Archive, learn (SG7)	1–2	3–6	1–2	Anonymisation and publication approval.
Total	**13–27**	**59–139**	**19–47**	

AL4 adds roughly 31 OPERATION claims and the analysis of a year of operating data: plan for a further 15–35 pilot-team and 5–15 host person-days. These too are assumptions.

PIL-REQ-008 Every report of pilot effort shall label the §8.2 figures as planning assumptions and shall report measured values separately.

9. Rules of engagement and safety

The rules of engagement are agreed in AIDCF-PIL-002 §7 (built from AIDCF-009 B14) and approved by the safety authority at SG2. In summary:

- The safety authority and capability owner may stop any pilot activity at any time; each stop is recorded with its effect on results (AMH-REQ-031).
- Assessors do not operate, switch, isolate, open or adjust any equipment (AMH-REQ-026). No test is performed (AMH-REQ-029 is not engaged because no test is authorised).
- Configuration extraction is read-only, scheduled with the system owner, and executed or witnessed under the host's change process where it touches production management interfaces. Extraction that the system owner judges could affect a controller or production workload is not performed; the item is recorded as an UNKNOWN Record (U2 inaccessible) or replaced by an acceptable alternative listed in the profile.
- Passive network discovery (named in some profiles, for example MCF-SEC-009) is used only where the host already operates it; the pilot does not deploy taps, sensors or scanners. Active scanning is prohibited.
- Site access follows the host's induction, escort, permit and personal protective equipment rules; walkthroughs follow an approved route (AIDCF-009 B10; Part B DT-10).
- Credentials issued for the pilot are named, read-only, time-limited and revoked at SG7; no credential is copied into workpapers (B14 step 4).

PIL-REQ-009 Any activity not listed as permitted in the signed rules of engagement is prohibited for the pilot team.

10. Roles and minimum competence

Roles are those of AIDCF-009 A4. Competence levels use AIDCF-009 Part B Part VII (AC-0 to AC-6); disciplines follow AIDCF-011 Part IX.

Role	Minimum competence for the mini-pilot	Separation
Sponsor (host)	Authority over the decision question, scope, resources and access.	Not independent reviewer.
Capability owner (host)	Accountable for the Capability Instance and outcomes; and each Variant B provider's owner.	Not assessor or reviewer for this Capability Instance.
Decision authority	AC-6: governance appointment, independence from the assessment team, recorded delegation (GC-13-6).	Not a preparer.
Lead assessor	AC-3: AIDCF method, evidence, gates and decision drafting; cross-domain integration.	Not independent reviewer or decision authority.
Domain assessors	AC-2 in their discipline. Collectively: electrical (power architecture, protection, UPS/generation); mechanical (cooling, liquid cooling, heat rejection); compute/storage/network (AI/HPC architecture, fabrics, storage tiers, recovery); OT/security (BMS/EPMS/PLC, zoning, control-plane authority); dependency and resilience. One person may cover more than one discipline where competent.	Not technical reviewer of own work.
Evidence lead	Provenance, integrity, classification and custody (AIDCF-008); runs the evidence manifest and the request list.	—
Technical reviewers	AC-4 per discipline reviewed.	Not preparer of the reviewed workpaper.
Independent reviewer	AC-5: impartial senior assurance competence; gate and decision challenge.	No preparation role; no prior design, operation or advisory work on the subject in the previous assessment period (APF-REQ-006).
Safety authority (host)	The site's appointed person for authorising access and work.	—
Research lead	Designs and analyses the research measures; may be the lead assessor only if the measures are reviewed by someone outside the team.	Research analysis never changes a result.
Second assessor (double assessment)	AC-2 in the discipline sampled; blind to the primary result until both are recorded.	Not the primary assessor for the same item.

A realistic minimum team is four to six people plus the independent reviewer and decision authority. Where a discipline is not covered, the affected results are NOT_ASSESSED or UNKNOWN and the limitation is disclosed (AMH-REQ-014). Licensed judgements (for example protection settings where regulation requires) are relied on only from an authorised professional, with reliance recorded (AMH-REQ-015).

PIL-REQ-010 Every assessor, reviewer and the research lead shall declare conflicts of interest in AIDCF-PIL-002 §13 before appointment.

11. Evidence handling and classification

11.1 Classification

Two classification schemes apply and are used together:

AIDCF-012 Part XIII (pilot records and reports)	SECURITY.md §3 (anything that may enter this repository)	Pilot use
Highly restricted	HIGHLY_RESTRICTED	Credentials, network addresses and paths, control-system configurations, privileged-access routes, physical routes, emergency procedures, exploitable weaknesses. Host systems only; never leaves the host's controlled repository.
Confidential	CONFIDENTIAL	Real architecture, evidence, findings, remediation and provider details. Controlled pilot repository; never published.
Client restricted	CONTROLLED (nearest equivalent)	Named-user reports (R2, R5) and identifiable metrics. Distribution list in AIDCF-PIL-002.
Public	PUBLIC	Anonymised, aggregated research results and an R8 Public Summary, only after host approval and classification review.

The two schemes are not identical (AIDCF-012 "Client restricted" is defined by named users; SECURITY.md "CONTROLLED" by repository approval). The mapping above is the pilot's working rule and is recorded as a framework observation.

AIDCF-PIL-003 assigns a default classification to each request row: HIGHLY_RESTRICTED for SEC and NET items and for IMPLEMENTATION evidence of CG-07 CRITICAL controls; CONFIDENTIAL otherwise. The evidence lead may raise, never lower, a default.

11.2 Handling

- Evidence is held in a controlled repository provided or approved by the host, with evidence ID, source, collector, custodian, classification and integrity reference (AMH-REQ-024; AIDCF-008).
- HIGHLY_RESTRICTED evidence is reviewed on host premises or host-controlled systems; workpapers cite it by evidence ID and summarise only what is needed for the result.
- Workpapers are version-controlled and never deleted (AMH-REQ-004).
- AI-assisted extraction, if used, is E7 derived evidence with lineage retained (AMH-REQ-028), runs only on host-approved systems, and is never independent corroboration of its own sources (GC-11-4).
- Retention and destruction follow AIDCF-PIL-002 §10.

PIL-REQ-011 No pilot evidence, workpaper or identifiable result shall be committed to this repository; only material classified PUBLIC under SECURITY.md after host approval may be.

12. Research measurement

12.1 Principle

Research measures are collected alongside the assessment and never feed back into a result, a Gate Result or the decision. The research lead owns them. They are recorded in two files:

- `pilots/pilot-results/control-claim-log-template.csv`: one row per mandatory control-claim (107 rows, generated), recording effort, evidence fulfilment, result, second-assessor result, UNKNOWN type and severity, and links to usability and defect items;
- `pilots/pilot-results/pilot-metrics-template.csv`: the summary measures PM-01 to PM-20 with definitions, formulas and sources.

Usability issues and defects are recorded in AIDCF-PIL-005.

12.2 Measures

ID	Measure	Definition
PM-01	Time per control-claim	Assessor hours per control-claim, median and inter-quartile range by domain and claim type.
PM-02	Host effort per evidence request	Host hours per fulfilled request.
PM-03	Evidence availability	Mandatory request rows fulfilled / issued; also fulfilled by due date.
PM-04	Above-AL3 evidence availability	AVAILABILITY_ONLY items that exist / requested; readiness indicator for AL4/AL5.
PM-05	UNKNOWN rate	UNKNOWN control-claim results / mandatory control-claims, with counts by type U1–U10 and severity US1–US5.
PM-06	NOT_ASSESSED rate	NOT_ASSESSED control-claims / mandatory control-claims, with cause.
PM-07	Sufficiency achieved	Distribution of achieved sufficiency S0–S5 against the applicable floor.
PM-08	Control-result agreement	Agreement between primary and second assessor on the double-assessment sample: raw agreement, the full contingency table, and Cohen's kappa (Cohen, 1960, "A coefficient of agreement for nominal scales") reported with its small-sample caveat.
PM-09	Gate Result agreement	Agreement on the nine evaluated Gate Results between the lead assessor and an independent second evaluation.
PM-10	Disagreement causes	Each disagreement classified: criterion ambiguity, evidence interpretation, scope or boundary, vocabulary, competence, recording error.
PM-11	Scope agreement	Differences between two independent first drafts of WP-02 and WP-03 (outcomes, tolerances, boundary items, dependencies) before reconciliation.
PM-12	Trace completeness	Material statements in the Decision Recommendation with a complete AMH-REQ-017 trace / all material statements.
PM-13	Usability issues	Count by severity and artefact (AIDCF-PIL-005).
PM-14	Framework defects	Defects in control requirements, assessment criteria, gate criteria, evidence profiles, vocabularies or templates, by type and artefact (AIDCF-PIL-005).
PM-15	Orphan findings	Material findings with no suitable control in the 360-control registry.
PM-16	Dependencies newly identified	Dependencies recorded under CG-03 that were absent from the host's own records before the pilot.
PM-17	Gate–score divergence	Domains or outcomes where the band would be B4/B5 while a relevant gate is FAIL or UNKNOWN, or the reverse.
PM-18	Decision usefulness	Named decision user's rating of relevance, clarity, actionability and trust in the output (AIDCF-PIL-004 §9), with free-text reasons.

ID	Measure	Definition
PM-19	Elapsed time and effort against plan	Actual against §8.2 assumptions, per phase.
PM-20	Interpretation questions	Cases where applying the standards needed an interpretation (for example §6.5, §6.6, §7.2), with the interpretation used and the result it affected.

12.3 Double assessment

At SG2 the research lead draws a stratified sample of about 20% of the 107 control-claims (at least one per in-scope gate's CRITICAL set and both claim types), using a recorded seed. A second assessor records an independent result for each sampled item from the same evidence package, before seeing the primary result. Disagreements are resolved by the technical reviewer through the normal process; the research record keeps both original results. All nine Gate Results are independently re-derived by a second competent person from the reviewed control results and criteria. Results from a single pilot are descriptive and are not used to claim reliability.

PIL-REQ-012 The primary and second assessor results shall both be retained unchanged in the control-claim log, whatever the resolved result.

13. Exit criteria

The pilot is complete when every item below holds. These incorporate the whitepaper §29.2 acceptance criteria.

#	Exit criterion
X1	SG0 to SG7 recorded with PROCEED (or STOP with the effect recorded, §13.1).
X2	Every mandatory control-claim has a WP-12 result in a canonical state; every assessed assertion has a stable identity and evidence reference.
X3	Every in-scope gate has a WP-14 Gate Result with claim depth, criteria status and ceiling; every NOT_ASSESSED gate is recorded with its ceiling.
X4	UNKNOWN Records, exclusions, conflicts and inaccessible scope remain visible in the decision and reports; CG-12 evaluated.
X5	Coverage calculated against the approved profile, and the 360-control figure shown for information.
X6	Independent review completed (WP-18) and the decision issued, or a recorded decision not to issue one (WP-19).
X7	No activity exceeded the signed rules of engagement, or each deviation is recorded and dispositioned.
X8	The issued decision is bounded by scope, states, period, evidence and intended users, in AIDCF-012 controlled wording.
X9	AIDCF-PIL-004 result record, AIDCF-PIL-005 lessons register and both metrics files completed.
X10	Credentials revoked, evidence retained or destroyed per AIDCF-PIL-002 §10, and destruction certified where required.
X11	Any public material has explicit written publication approval from the host.

13.1 Early stop

The pilot stops, with the stop recorded at the current stage gate, if an unsafe condition arises, authority becomes unresolved, source integrity fails, scope is breached, the safety authority or capability owner exercises the stop right, or a sponsor withdraws. Work completed remains in the record. If the stop occurs before SG6 no decision state is recorded, and any report uses the AIDCF-012 V.3 wording. A stopped pilot still produces AIDCF-PIL-004 and AIDCF-PIL-005, and its measures are reported as a stopped case.

PIL-REQ-013 A stopped or incomplete pilot shall be reported as such and shall not be presented as a completed pilot.

14. Publication and feedback into AIDCF

14.1 Publication

- Nothing is published without the host's written approval under AIDCF-PIL-002 §12, given after the host has seen the exact text.
- Published research outputs are anonymised and aggregated: no host name unless the host asks to be named; no site, location, supplier, product, network or topology details; no detail that would let a reader identify a single point of failure or weakness (SECURITY.md §4).
- Small counts that could identify a condition at a recognisable site are suppressed or banded.
- The Assurance Decision itself is published only as an R8 Public Summary, in AIDCF-012 controlled wording with all mandatory disclosures, and only if the host approves. Redaction never removes the decision state, controlling Gate Results, US4/US5 UNKNOWN Records or material limitations (RAS-REQ-012).
- Negative and null results are published on the same terms as positive ones (README research questions).

14.2 Repository records

After host approval, the following may be added to this repository, each classification-reviewed as PUBLIC: the completed AIDCF-PIL-004 with identifying fields redacted to the agreed level; the anonymised metrics file; the AIDCF-PIL-005 register with host-specific content removed; and change proposals. The whitepaper §29.4 record and README pilot status are updated only by the framework owner, and only on the basis of those signed records.

14.3 Feedback

Each AIDCF-PIL-005 lesson that implies a change to an AIDCF artifact becomes a change proposal in the GOVERNANCE.md §9 format, classified editorial, minor normative, major normative or constitutional (GOVERNANCE.md §8). Registry defects (control requirements, criteria, profiles, gate mappings) go to the AIDCF-007A amendment process. Method and template defects go to the owners of AIDCF-009, AIDCF-010, AIDCF-011 or AIDCF-012. The pilot team proposes; the framework owner and reviewers decide.

PIL-REQ-014 Every PM-14 defect and PM-20 interpretation question shall be disposed of in AIDCF-PIL-005 as a change proposal, a deferral with reason, or a rejection with reason.

PIL-REQ-015 Pilot outputs shall not use the claims prohibited by AIDCF-011 Part XIV or GOVERNANCE.md §17, including "AIDCF Certified", "externally validated" or "production proven".

PIL-REQ-016 Each assurance statement or report issued from the pilot shall carry the AIDCF-012 non-certification note.

15. Regenerating the machine-readable files

```
python3 pilots/pilot-framework/tools/build_starter_profile.py           # write
python3 pilots/pilot-framework/tools/build_starter_profile.py --check   # verify; exit 1 if
stale
```

The generator uses the Python standard library only, reads `registries/controls/controls.csv`, `evidence-profiles.csv`, `gate-mappings.csv`, `registries/gates/critical-gates.csv`, `critical-gate-criteria.csv`, `registries/domains/domains.csv` and `registries/vocabularies/claim-types.csv`, and writes the Starter Profile, the Evidence Request List and the control-claim log template. Run `--check` after any registry change; if it fails, regenerate, review the difference, and record the change at the current stage gate if a pilot is under way.